R18

Code No: 157GD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech IV Year I Semester Examinations, June/July - 2024

DATABASE SECURITY

(Computer Science and Engineering – Data Science)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) What is meant by hostile agent? [2]

b) List the three main aspects of information security in a database. [3]

c) How to construct secure passwords? [2]

d) Give low-watermark policy for subjects in Biba model. [3]

e) What is a well-formed transaction? [2]

f) Define partial compromise. [3]

g) What is an aggregate class? [2]

h) Give an example of role lattice. [3]

i) What is a table function? [2]

j) Give the syntax of MLS rule. [3]

PART – B

(50 Marks)

2. Elucidate various threats to database security. [10]

OR

3.a) Discuss the mutual consistency rules for static graph and dynamic graph of Acten model.

b) What are the challenges in securing distributed databases? [5+5]

4. Make a comparison of Bell-LaPadula Model and Access matrix model. [10]

OR

5. Describe any three types of hardware mechanisms for protection and controlled memory sharing. [10]

6. Compare and contrast kernelized architecture with replicated architecture. [10]

OR

7. With the help of a diagram describe the architecture of RETISS system. [10]

8. Demonstrate the role of content constraints and context constraints in SORION model. [10]

OR

9. Differentiate between discretionary security models and mandatory protection for object-oriented databases. [10]

10. What are the various security attacks and security mechanisms for active database? Explain with example. [10]

OR

11. With the help of a neat sketch, describe ORION model for protection of active database.[10]

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